How-To Library

Informatica

Configuring Ports for Enterprise Information Catalog

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Abstract

You can deploy an Enterprise Data Catalog solution to address your requirements to extract, process and report data and metadata from big data sources. To prevent conflicts between ports, you need to know the recommended values for each port and default port numbers. This article lists the ports that are established when you run the installer for Enterprise Data Catalog.

Supported Versions

- Informatica Enterprise Information Catalog 10.2

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Overview

You can configure the Enterprise Data Catalog solution to extract, process and report metadata.

When you install Enterprise Data Catalog, the installer utility opens specific ports by default. You must open other ports manually. This article tells you which ports are established when you run the installer.

Note: All ports need to be bidirectional.

Informatica Domain and Application Services

The Informatica domain includes several services that perform important roles in metadata extraction and processing.

For more information about application services, see the Informatica 10.4.0 Application Service Guide.

Application Services and Ports

Informatica domain services and application services in the Informatica domain have unique ports.

Informatica Domain

The following table describes the ports that you can set:

Port	Description
Service Manager port	Port number used by the Service Manager on the node. The Service Manager listens for incoming connection requests on this port. Client applications use this port to communicate with the services in the domain. The Informatica command line programs use this port to communicate to the domain. This is also the port for the SQL data service JDBC/ODBC driver. Default is 6006.
Service Manager Shutdown port	Port number that controls server shutdown for the domain Service Manager. The Service Manager listens for shutdown commands on this port. Default is 6007.
Informatica Administrator port	Port number used by Informatica Administrator. Default is 6008.
Informatica Administrator shutdown port	Port number that controls server shutdown for Informatica Administrator. Informatica Administrator listens for shutdown commands on this port. Default is 6009.
Minimum port number	Lowest port number in the range of dynamic port numbers that can be assigned to the application service processes that run on this node. Default is 6014.
Maximum port number	Highest port number in the range of dynamic port numbers that can be assigned to the application service processes that run on this node. Default is 6114.

Analyst Service

The following table lists the default port associated with the Analyst Service:

Type	Default Port
Analyst Service (HTTP)	8085
Analyst Service (HTTPS)	No default port. Enter the required port number when you create the service.
Analyst Service (Staging database)	No default port. Enter the database port number.

Content Management Service

The following table lists the default port associated with the Content Management Service:

Type	Default Port
Content Management Service (HTTP)	8105
Content Management Service (HTTPS)	No default port. Enter the required port number when you create the service.

Data Integration Service

The following table lists the default port associated with the Data Integration Service:

Type	Default Port
Data Integration Service (HTTP proxy)	8085
Data Integration Service (HTTP)	8095
Data Integration Service (HTTPS)	No default port. Enter the required port number when you create the service.
Profiling Warehouse database	No default port. Enter the database port number.
Human Task database	No default port. Enter the database port number.

Enterprise Data Catalog Ports

The following table lists ports that the installer opens when you install Enterprise Data Catalog:

Description	Port Number
Informatica Administrator	- HTTP 6008 - HTTPS 8443
Catalog Service	9085
Informatica Cluster Service	9075

Hadoop Cluster Ports

Make sure that each node in the Hadoop cluster can access all the ports of the other nodes in the cluster.

The following table lists the Hadoop cluster ports used by applications launched on the cluster by Enterprise Data Catalog:

Description	Port Number
HTTPS Port for Hadoop Distributions	- Cloudera 7183 - Hortonworks 8443 - Azure HDInsight 8443
Ambari Agent TCP	8670
Ambari Metrics Collector TCP	6188, 60200
Ambari Server TCP	8080, 8440, 8441
Ambari Server UDP	59804
HDFS DataNode	50010
HDFS DataNode HTTP	50075

Description	Port Number
HDFS DataNode HTTPS	50475
HDFS DataNode IPC	8010
HDFS DFSZKFailoverController	8019
HDFS HTTPS	50470
HDFS JournalNode HTTP	8480
HDFS JournalNode HTTPS	8481
HDFS NameNode HTTP	50070
HDFS NameNode HTTPS	50470
HDFS NameNode RPC	8020
HDFS Secondary Name Node	50090
pgSQL	5432
YARN JobHistory Server	19888
YARN NodeManager	45454
YARN ResourceManager	8032
YARN ResourceManager Admin	8141
YARN ResourceManager ResourceTracker	8025
YARN ResourceManager Scheduler	8030
YARN ResourceManager Webapp	8088
YARN ResourceManager Webapp HTTPS	8090
YARN Timeline Server	10020
YARN Timeline Server Webapp	8188
YARN Timeline Server Webapp HTTPS	8190
Zookeeper	2181

Note: You can control the RPC ports of the YARN applications, HBase and Solr, launched by Enterprise Data Catalog on the Hadoop cluster by specifying a range in the following properties:

- LdmCustomOptions.deployment.slider.max.port
- LdmCustomOptions.deployment.slider.min.port

Big Data Management Ports

The following sections list Hadoop components and default port numbers for various Hadoop distributions:

IBM BigInsights

The following table lists the default ports associated with IBM BigInsights:

Hadoop Component	Port
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9933
HiveServer	10000
JobTracker	9001
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	9000
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTP	13562
TaskTracker web management	50060
ZooKeeper	2181

Cloudera Port Reference

The following table lists the ports associated with Cloudera:

Hadoop Component	Port
Cloudera Manager	7180
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9083
HiveServer	10000
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	8020
ResourceManager	8050
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTP	13562
TaskTracker web management	50060
ZooKeeper	2181

HDInsight Port Reference

The following table lists the ports associated with HDInsight:

Hadoop Component	Port
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9933
HiveServer	10000
JobTracker	9001
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	9000
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTP	13562
TaskTracker web management	50060
ZooKeeper	2181

Hortonworks Port Reference

The following table lists the ports associated with Hortonworks:

Hadoop Component	Port
HBase master	60000
HBase master web	60010

Hadoop Component	Port
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9933
HiveServer	10000
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	8020
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTP	13562
TaskTracker web management	50060
ZooKeeper	2181

MapR Port Reference

The following table lists the ports associated with MapR:

Hadoop Component	Port	Notes
CLDB	7222	
CLDB JMX monitor	7220	
CLDB web port	7221	
HBase master	16000	
HBase master web	16010	

Hadoop Component	Port	Notes
HBase region server	16020	
Hive metastore	9083	
HiveServer	10000	
JobTracker	9001	
JobTracker web manager	50030	
MapReduce Application Master	50100-50200	
MapReduce JobHistory server	10020	YARN only
MapReduce JobHistory server web app	19888	YARN only
MySQL	3306	Required if you use MySQL.
ResourceManager	8032	YARN only
ResourceManager web app	8088	YARN only
Secure Resource Manager web app (HTTPS)	8090	
ResourceTracker	8031	YARN only
Scheduler address	8030	YARN only
Shuffle HTP	13562	
TaskTracker web management	50060	
ZooKeeper	5181	

Hadoop Cluster Port Reference

A Hadoop cluster requires some unique port numbers.

Blaze Engine Port Reference

The Blaze engine uses ports for the Blaze Job Monitor and the Blaze services.

Blaze Job Monitor

The following table lists the ports for the Blaze Job Monitor:

Port Name	Default Port Number
HTTP	9080
JSF	9090

Blaze Services

The following table lists the ports for Blaze services:

Port Name	Default Port Number
Minimum and Maximum Port Range for Hadoop Connection	12300 - 12600

Developer Tool Port Reference

The following table lists the ports for the Developer tool:

Port Name	Default Port Number
CLDB	7222 - MapR only
HBase master	60000
HBase region server	60200
HiveServer2 thrift	10000
NameNode RPC	8020 - except MapR
ZooKeeper client	2181, 5181

Default Ports for Source Systems

The following table lists the default ports for various source systems from which you extract metadata into Enterprise Data Catalog:

Source System	Default Port Number
Amazon Redshift	5439
Amazon S3	n/a
Atlas	21000
Business Glossary	8085, the default port of the Analyst Service.
Cloudera Navigator	7180
ERwin	18170
Hive	10000
HDFS	8020
IBM Cognos	9300
IBM DB2	50000

Source System	Default Port Number
IBM DB2 for z/OS	446
IBM Netezza	5480
Informatica Axon	9443 for SSL domain and 9999 for non-SSL domain.
Informatica Cloud Service	n/a
Informatica Platform	6005, the default port of Informatica domain.
Informatica PowerCenter	6005, the default port of Informatica domain.
Microsoft Azure SQL Data Warehouse	1433
Microsoft Azure SQL Database	1433
MicroStrategy	n/a
Oracle	1521
Oracle Business Intelligence Enterprise Edition	7001
SAP Business Objects	6400
SAP R/3	n/a
SQL Server	1433
SQL Server Integration Services (SSIS)	n/a
Salesforce	n/a
Sybase ASE	5000
Tableau Server	8000
Teradata	1025

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